# Global Correlation Experiment — Analysis Report

Can LLM relevance scoring replace or supplement the retriever when it fails?

| RUN | MODEL | BACKEND | SCORED PAIRS | QUERIES | STRIDE | GENERATED |
|---|---|---|---|---|---|---|
| staged_run_2026-05-17_18-18-42 | gcp/google/gemini-2.5-flash-lite | nvidia_ih | 1,004,140 | 199 | 10 | 2026-06-12 13:37 |

## About This Analysis

> **What is this experiment?** A RAG (Retrieval-Augmented Generation) pipeline depends on a vector retriever to surface the right document before a language model answers. When the retriever fails — i.e. the ground-truth document (GT) is not ranked first — the downstream answer quality drops. This experiment tests whether an LLM relevance score can identify the GT document from a large candidate pool and rescue those failed retrievals.

### RESEARCH QUESTION

*When the retriever does not rank the GT document at position 1, can an LLM relevance score re-rank the pool and recover it?*

### HOW THE EXPERIMENT WORKS

For each of the 199 queries a document pool of up to 6 000 candidates is assembled — a mix of embedding-close and embedding-distant documents. Every `(query, document)` pair is then scored by an LLM on an integer 1–10 scale (stored as 0.1–1.0). The total scored pairs in this run: **1,004,140**.

Each row in the database carries four columns the analysis relies on:

| Column | What it is | Role in analysis |
|---|---|---|
| `llm_score` | LLM relevance score (0.1 – 1.0) | The signal we are testing as a re-ranker |
| `dist_to_gt` | Cosine distance from this doc's embedding to the GT embedding | Ground-truth proximity proxy; GT doc itself has dist ≈ 0 |
| `is_gt` | 1 if this doc is the ground truth for this query, else 0 | Label for binary classification metrics (ROC, P@K) |
| `rag_failed` | 1 if the retriever did not rank GT at position 1 | Scope flag — all analyzed queries have rag_failed = 1 |

### WHAT THE ANALYSIS MEASURES — AND WHY EACH PART MATTERS

| Analysis part | Script | Core question answered |
|---|---|---|
| **Base analysis**<br>*15 sections* | `analyze_experiment.py` | Does LLM score correlate with distance to GT? How often does it rank GT near the top? Produces correlation coefficients, ROC/PR curves, recovery@K, score distributions. |
| **Extended / corrective analysis**<br>*10 sections A–J* | `analyze_experiment_extended.py` | Fixes misleading conclusions from the base run. The LLM uses a 1–10 integer scale — many docs share the same top score, so "P@1 = 98%" is inflated by ties. This part surfaces the tie structure, corrects MRR, classifies query difficulty, and tests whether embedding distance can break ties within the top-scoring group. |
| **HTML report** | `export_report.py` | Assembles all charts, tables, and narrative into this self-contained file. |

### KEY CONCEPT: THE TIE PROBLEM

> ⚠ **Why "P@1 = 98%" is not the right headline number.**
> The LLM scores documents on a 1–10 integer scale (÷10), producing only 10 distinct values. In most queries, both the GT document *and* ~9 other documents all receive the maximum score of 1.0. Ranking them all at position 1 makes GT appear uniquely identified — but it is not. The tie-corrected MRR (0.26, not 0.98) is the honest estimate of LLM-as-ranker accuracy. The extended analysis sections A and G explain this in detail.

### THE ACTIONABLE FINDING

> ★ **Two-stage hybrid pipeline: LLM filter → embedding-distance tiebreaker.**
> At threshold 1.0 the LLM reduces the candidate pool from ~6 000 to ~11 documents while retaining GT in 98% of queries. Within that small pool, the GT document has the lowest cosine distance to the GT embedding in **100% of applicable queries**. Combining both signals achieves near-perfect retrieval on this dataset.

## Executive Summary

⚠ **Partial run — metrics reflect scored data only; see details below.** **Status:** Incomplete **Synced:** 1,004,140 / 1,004,400 rows (100.0%) **Queries:** 199 analyzed (1 skipped, 199 in DB) **Doc pool:** 4,000–6,000 docs/query (95 queries partial) **Last stage:** 2 **Fatal error (query 167):** 401 Client Error: Unauthorized for url: https://inference-api.nvidia.com/vertex_ai/v1/projects/n20250411173346/locations/global/pub

| 0.9960 | -0.166 | 0.257 | 98.0% | 99.0% | 9 |
|---|---|---|---|---|---|
| ROC-AUC | SPEARMAN R | MRR (TIE-CORRECTED) | RECOVERY@1 | RECOVERY@10 | MEDIAN TIE SIZE |
| binary GT detection | score vs dist_to_gt | optimistic: 0.981 | GT in LLM top-1 | GT in LLM top-10 | docs tied at GT's score |

| 0.172 | 100.0% |
|---|---|
| TIER PRECISION | HYBRID WIN RATE |
| expected if random pick | LLM + dist tiebreaker |

★ The retriever failed on **100%** of analyzed queries. Despite this, the LLM assigns the maximum relevance score (1.0) to the GT document in **98.0%** of queries — but it also assigns 1.0 to a median of **9** other documents, so it acts as a **coarse filter** rather than a precise ranker. Adding an embedding-distance tiebreaker resolves ties in **100.0%** of top-tier queries (95.0% overall).

### QUERY DIFFICULTY BREAKDOWN

| Category | Queries | Definition |
|---|---|---|
| EASY | 6 | GT is the only doc scoring 1.0 |
| MEDIUM | 100 | GT scores 1.0, 2–10 others also do |
| HARD | 89 | GT scores 1.0, >10 others also do |
| FAILED | 4 | GT does not reach score 1.0 |

## Reading the Outputs

Analysis outputs are split by **scope** so you can tell which numbers describe the whole experiment vs a single query:

- **Global** — one summary over all queries and doc pairs (e.g. MRR, Recovery@K, ROC-AUC). Files in `csv/global/`.
- **Per-query** — one row per `query_idx` (e.g. GT rank, difficulty, tie size). Files in `csv/per_query/`. Filter in Excel/Sheets to inspect specific queries.

ℹ **Start per-query review:** open `csv/per_query/per_query_overview.csv` — merged master table for all 199 queries. See also `INDEX.md` in the analysis folder.

### PER-QUERY OVERVIEW (ALL QUERIES)

### COLUMN GUIDE

| Column | Meaning |
|---|---|
| `query_idx` | Unique integer ID for the query (0-based) |

| Column | Meaning |
|---|---|
| `gt_llm_rank` | Rank of the ground-truth document by LLM score (1 = top-scored). Lower is better. |
| `gt_llm_score` | Raw LLM relevance score assigned to the GT document (1–10 scale, normalised to 0.1–1.0) |
| `difficulty` | Query tier: EASY = GT uniquely top-scored · MEDIUM = 2–10 docs tied with GT · HARD = 11+ tied · FAILED = GT outscored |
| `n_tied_at_gt` | Number of documents that share the same LLM score as the GT doc (including GT itself). Large values indicate the LLM could not discriminate. |
| `rank_avg` | Tie-corrected average rank of the GT doc: (rank_min + rank_max) / 2. Accounts for ties; lower = better. |
| `spearman_r` | Spearman rank correlation between LLM score and distance-to-GT for this query (range –1 to +1). Strong negative = LLM score tracks proximity well. |
| `gt_would_win_dist` | 1 if the GT doc has a shorter embedding distance to itself (trivially 0) than the closest non-GT in the same score tier — i.e. whether a distance tiebreaker would promote GT to rank 1. 0 otherwise. |

| query_idx | gt_llm_rank | gt_llm_score | difficulty | n_tied_at_gt | rank_avg | spearman_r | gt_would_win_dist |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.096 | 1.0 |
| 2 | 1 | 1.0 | HARD | 18 | 9.5 | -0.190 | 1.0 |
| 3 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.091 | 1.0 |
| 4 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.086 | 1.0 |
| 5 | 1 | 1.0 | HARD | 34 | 17.5 | -0.220 | 1.0 |
| 6 | 1 | 1.0 | HARD | 12 | 6.5 | -0.108 | 1.0 |
| 7 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.345 | 1.0 |
| 8 | 1 | 1.0 | HARD | 32 | 16.5 | -0.125 | 1.0 |
| 9 | 1 | 1.0 | HARD | 14 | 7.5 | -0.250 | 1.0 |
| 10 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.215 | 1.0 |
| 11 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.098 | 1.0 |
| 12 | 1 | 1.0 | HARD | 14 | 7.5 | -0.218 | 1.0 |
| 13 | 1 | 1.0 | HARD | 13 | 7.0 | -0.193 | 1.0 |
| 14 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.266 | 1.0 |
| 15 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.151 | 1.0 |
| 16 | 1 | 1.0 | MEDIUM | 6 | 3.5 | -0.114 | 1.0 |
| 17 | 1 | 1.0 | HARD | 14 | 7.5 | -0.122 | 1.0 |
| 18 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.132 | 1.0 |
| 19 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.137 | 1.0 |
| 20 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.347 | 1.0 |
| 21 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.178 | 1.0 |
| 22 | 1 | 1.0 | HARD | 11 | 6.0 | -0.227 | 1.0 |
| 23 | 1 | 1.0 | EASY | 1 | 1.0 | -0.275 | nan |
| 24 | 1 | 1.0 | HARD | 52 | 26.5 | -0.175 | 1.0 |
| 25 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.144 | 1.0 |
| 26 | 1 | 1.0 | HARD | 25 | 13.0 | -0.134 | 1.0 |
| 27 | 1 | 1.0 | HARD | 30 | 15.5 | -0.114 | 1.0 |
| 28 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.184 | 1.0 |
| 29 | 1 | 1.0 | HARD | 11 | 6.0 | -0.088 | 1.0 |
| 30 | 1 | 1.0 | HARD | 12 | 6.5 | -0.174 | 1.0 |
| 31 | 1 | 1.0 | HARD | 17 | 9.0 | -0.055 | 1.0 |
| 32 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.064 | 1.0 |
| 33 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.118 | 1.0 |
| 34 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.056 | 1.0 |
| 35 | 1 | 1.0 | HARD | 19 | 10.0 | -0.194 | 1.0 |
| 36 | 1 | 1.0 | HARD | 14 | 7.5 | -0.297 | 1.0 |
| 37 | 1 | 1.0 | HARD | 29 | 15.0 | -0.166 | 1.0 |
| 38 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.138 | 1.0 |
| 39 | 1 | 1.0 | HARD | 19 | 10.0 | -0.108 | 1.0 |

| query_idx | gt_llm_rank | gt_llm_score | difficulty | n_tied_at_gt | rank_avg | spearman_r | gt_would_win_dist |
|---|---|---|---|---|---|---|---|
| 40 | 1 | 1.0 | HARD | 20 | 10.5 | -0.223 | 1.0 |
| 41 | 1 | 1.0 | HARD | 14 | 7.5 | -0.117 | 1.0 |
| 42 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.204 | 1.0 |
| 43 | 6 | 0.8 | FAILED | 9 | 10.0 | -0.161 | nan |
| 44 | 1 | 1.0 | HARD | 22 | 11.5 | -0.276 | 1.0 |
| 45 | 1 | 1.0 | HARD | 18 | 9.5 | -0.189 | 1.0 |
| 46 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.141 | 1.0 |
| 47 | 1 | 1.0 | HARD | 16 | 8.5 | -0.066 | 1.0 |
| 48 | 1 | 1.0 | HARD | 16 | 8.5 | -0.089 | 1.0 |
| 49 | 1 | 1.0 | EASY | 1 | 1.0 | -0.205 | nan |
| 50 | 1 | 1.0 | HARD | 29 | 15.0 | -0.238 | 1.0 |
| 51 | 1 | 1.0 | MEDIUM | 10 | 5.5 | -0.205 | 1.0 |
| 52 | 1 | 1.0 | HARD | 12 | 6.5 | -0.124 | 1.0 |
| 53 | 1 | 1.0 | HARD | 26 | 13.5 | -0.195 | 1.0 |
| 54 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.226 | 1.0 |
| 55 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.134 | 1.0 |
| 56 | 1 | 1.0 | HARD | 12 | 6.5 | -0.205 | 1.0 |
| 57 | 1 | 1.0 | HARD | 13 | 7.0 | -0.217 | 1.0 |
| 58 | 1 | 1.0 | HARD | 25 | 13.0 | -0.129 | 1.0 |
| 59 | 1 | 1.0 | HARD | 11 | 6.0 | -0.165 | 1.0 |
| 60 | 1 | 1.0 | HARD | 15 | 8.0 | -0.097 | 1.0 |
| 61 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.041 | 1.0 |
| 62 | 1 | 1.0 | HARD | 18 | 9.5 | -0.279 | 1.0 |
| 63 | 1 | 1.0 | HARD | 14 | 7.5 | -0.194 | 1.0 |
| 64 | 1 | 1.0 | MEDIUM | 6 | 3.5 | -0.067 | 1.0 |
| 65 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.100 | 1.0 |
| 66 | 1 | 1.0 | HARD | 24 | 12.5 | -0.058 | 1.0 |
| 67 | 1 | 1.0 | HARD | 12 | 6.5 | -0.162 | 1.0 |
| 68 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.195 | 1.0 |
| 69 | 1 | 1.0 | HARD | 12 | 6.5 | -0.110 | 1.0 |
| 70 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.208 | 1.0 |
| 71 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.102 | 1.0 |
| 72 | 1 | 1.0 | HARD | 24 | 12.5 | -0.122 | 1.0 |
| 73 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.260 | 1.0 |
| 74 | 1 | 1.0 | HARD | 15 | 8.0 | -0.149 | 1.0 |
| 75 | 1 | 1.0 | HARD | 26 | 13.5 | -0.148 | 1.0 |
| 76 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.233 | 1.0 |
| 77 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.289 | 1.0 |
| 78 | 1 | 1.0 | HARD | 15 | 8.0 | -0.117 | 1.0 |
| 79 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.235 | 1.0 |
| 80 | 1 | 1.0 | HARD | 16 | 8.5 | -0.152 | 1.0 |
| 81 | 1 | 1.0 | HARD | 16 | 8.5 | -0.079 | 1.0 |
| 82 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.110 | 1.0 |
| 83 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.168 | 1.0 |
| 84 | 1 | 1.0 | HARD | 16 | 8.5 | -0.168 | 1.0 |
| 85 | 1 | 1.0 | MEDIUM | 3 | 2.0 | -0.053 | 1.0 |
| 86 | 1 | 1.0 | MEDIUM | 3 | 2.0 | -0.209 | 1.0 |
| 87 | 28 | 0.6 | FAILED | 39 | 47.0 | -0.129 | nan |
| 88 | 1 | 1.0 | EASY | 1 | 1.0 | -0.090 | nan |
| 89 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.171 | 1.0 |

| query_idx | gt_llm_rank | gt_llm_score | difficulty | n_tied_at_gt | rank_avg | spearman_r | gt_would_win_dist |
|---|---|---|---|---|---|---|---|
| 90 | 1 | 1.0 | HARD | 18 | 9.5 | -0.178 | 1.0 |
| 91 | 1 | 1.0 | HARD | 13 | 7.0 | -0.194 | 1.0 |
| 92 | 1 | 1.0 | MEDIUM | 6 | 3.5 | -0.139 | 1.0 |
| 93 | 1 | 1.0 | HARD | 15 | 8.0 | -0.266 | 1.0 |
| 94 | 1 | 1.0 | HARD | 30 | 15.5 | -0.225 | 1.0 |
| 95 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.138 | 1.0 |
| 96 | 1 | 1.0 | HARD | 15 | 8.0 | -0.092 | 1.0 |
| 97 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.101 | 1.0 |
| 98 | 1 | 1.0 | MEDIUM | 6 | 3.5 | -0.298 | 1.0 |
| 99 | 1 | 1.0 | HARD | 19 | 10.0 | -0.075 | 1.0 |
| 100 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.225 | 1.0 |
| 101 | 1 | 1.0 | HARD | 14 | 7.5 | -0.222 | 1.0 |
| 102 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.136 | 1.0 |
| 103 | 1 | 1.0 | HARD | 20 | 10.5 | -0.119 | 1.0 |
| 104 | 1 | 1.0 | HARD | 28 | 14.5 | -0.221 | 1.0 |
| 105 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.113 | 1.0 |
| 106 | 1 | 1.0 | HARD | 18 | 9.5 | -0.208 | 1.0 |
| 107 | 281 | 0.1 | FAILED | 3720 | 2140.5 | 0.010 | nan |
| 108 | 1 | 1.0 | MEDIUM | 3 | 2.0 | -0.278 | 1.0 |
| 109 | 1 | 1.0 | HARD | 12 | 6.5 | -0.249 | 1.0 |
| 110 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.273 | 1.0 |
| 111 | 1 | 1.0 | EASY | 1 | 1.0 | -0.239 | nan |
| 112 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.135 | 1.0 |
| 113 | 1 | 1.0 | HARD | 15 | 8.0 | -0.107 | 1.0 |
| 114 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.278 | 1.0 |
| 115 | 1 | 1.0 | HARD | 11 | 6.0 | -0.132 | 1.0 |
| 116 | 1 | 1.0 | MEDIUM | 3 | 2.0 | -0.150 | 1.0 |
| 117 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.250 | 1.0 |
| 118 | 1 | 1.0 | MEDIUM | 10 | 5.5 | -0.281 | 1.0 |
| 119 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.120 | 1.0 |
| 120 | 1 | 1.0 | HARD | 18 | 9.5 | -0.156 | 1.0 |
| 121 | 1 | 1.0 | MEDIUM | 10 | 5.5 | -0.142 | 1.0 |
| 122 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.078 | 1.0 |
| 123 | 1 | 1.0 | HARD | 15 | 8.0 | -0.246 | 1.0 |
| 124 | 1 | 1.0 | HARD | 19 | 10.0 | -0.325 | 1.0 |
| 125 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.241 | 1.0 |
| 126 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.183 | 1.0 |
| 127 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.126 | 1.0 |
| 128 | 1 | 1.0 | HARD | 17 | 9.0 | -0.166 | 1.0 |
| 129 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.130 | 1.0 |
| 130 | 1 | 1.0 | HARD | 13 | 7.0 | -0.250 | 1.0 |
| 131 | 1 | 1.0 | HARD | 14 | 7.5 | -0.294 | 1.0 |
| 132 | 1 | 1.0 | HARD | 11 | 6.0 | -0.200 | 1.0 |
| 133 | 1 | 1.0 | MEDIUM | 3 | 2.0 | -0.151 | 1.0 |
| 134 | 1 | 1.0 | MEDIUM | 10 | 5.5 | -0.300 | 1.0 |
| 135 | 1 | 1.0 | HARD | 30 | 15.5 | -0.321 | 1.0 |
| 136 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.197 | 1.0 |
| 137 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.166 | 1.0 |
| 138 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.190 | 1.0 |
| 139 | 1 | 1.0 | HARD | 13 | 7.0 | -0.217 | 1.0 |

| query_idx | gt_llm_rank | gt_llm_score | difficulty | n_tied_at_gt | rank_avg | spearman_r | gt_would_win_dist |
|---|---|---|---|---|---|---|---|
| 140 | 9 | 0.8 | FAILED | 10 | 13.5 | -0.189 | nan |
| 141 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.164 | 1.0 |
| 142 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.162 | 1.0 |
| 143 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.112 | 1.0 |
| 144 | 1 | 1.0 | MEDIUM | 6 | 3.5 | -0.102 | 1.0 |
| 145 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.118 | 1.0 |
| 146 | 1 | 1.0 | HARD | 23 | 12.0 | -0.157 | 1.0 |
| 147 | 1 | 1.0 | MEDIUM | 3 | 2.0 | -0.285 | 1.0 |
| 148 | 1 | 1.0 | HARD | 14 | 7.5 | -0.104 | 1.0 |
| 149 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.253 | 1.0 |
| 150 | 1 | 1.0 | HARD | 23 | 12.0 | -0.153 | 1.0 |
| 151 | 1 | 1.0 | MEDIUM | 10 | 5.5 | -0.265 | 1.0 |
| 152 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.235 | 1.0 |
| 153 | 1 | 1.0 | HARD | 23 | 12.0 | -0.235 | 1.0 |
| 154 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.228 | 1.0 |
| 155 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.303 | 1.0 |
| 156 | 1 | 1.0 | HARD | 11 | 6.0 | -0.222 | 1.0 |
| 157 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.256 | 1.0 |
| 158 | 1 | 1.0 | HARD | 18 | 9.5 | -0.273 | 1.0 |
| 159 | 1 | 1.0 | HARD | 12 | 6.5 | -0.131 | 1.0 |
| 160 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.239 | 1.0 |
| 161 | 1 | 1.0 | HARD | 11 | 6.0 | -0.241 | 1.0 |
| 162 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.122 | 1.0 |
| 163 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.142 | 1.0 |
| 164 | 1 | 1.0 | HARD | 22 | 11.5 | -0.144 | 1.0 |
| 165 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.331 | 1.0 |
| 166 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.139 | 1.0 |
| 167 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.119 | 1.0 |
| 168 | 1 | 1.0 | HARD | 11 | 6.0 | -0.224 | 1.0 |
| 169 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.095 | 1.0 |
| 170 | 1 | 1.0 | HARD | 14 | 7.5 | -0.063 | 1.0 |
| 171 | 1 | 1.0 | HARD | 16 | 8.5 | -0.174 | 1.0 |
| 172 | 1 | 1.0 | HARD | 15 | 8.0 | -0.135 | 1.0 |
| 173 | 1 | 1.0 | MEDIUM | 6 | 3.5 | -0.211 | 1.0 |
| 174 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.105 | 1.0 |
| 175 | 1 | 1.0 | HARD | 13 | 7.0 | -0.355 | 1.0 |
| 176 | 1 | 1.0 | MEDIUM | 6 | 3.5 | -0.144 | 1.0 |
| 177 | 1 | 1.0 | HARD | 13 | 7.0 | -0.236 | 1.0 |
| 178 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.279 | 1.0 |
| 179 | 1 | 1.0 | MEDIUM | 7 | 4.0 | -0.177 | 1.0 |
| 180 | 1 | 1.0 | EASY | 1 | 1.0 | -0.078 | nan |
| 181 | 1 | 1.0 | MEDIUM | 6 | 3.5 | -0.135 | 1.0 |
| 182 | 1 | 1.0 | MEDIUM | 3 | 2.0 | -0.129 | 1.0 |
| 183 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.082 | 1.0 |
| 184 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.328 | 1.0 |
| 185 | 1 | 1.0 | HARD | 14 | 7.5 | -0.238 | 1.0 |
| 186 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.136 | 1.0 |
| 187 | 1 | 1.0 | EASY | 1 | 1.0 | -0.151 | nan |
| 188 | 1 | 1.0 | MEDIUM | 2 | 1.5 | -0.263 | 1.0 |
| 189 | 1 | 1.0 | MEDIUM | 4 | 2.5 | -0.147 | 1.0 |

| query_idx | gt_llm_rank | gt_llm_score | difficulty | n_tied_at_gt | rank_avg | spearman_r | gt_would_win_dist |
|---|---|---|---|---|---|---|---|
| 190 | 1 | 1.0 | HARD | 40 | 20.5 | -0.164 | 1.0 |
| 191 | 1 | 1.0 | MEDIUM | 8 | 4.5 | -0.235 | 1.0 |
| 192 | 1 | 1.0 | HARD | 15 | 8.0 | -0.086 | 1.0 |
| 193 | 1 | 1.0 | HARD | 15 | 8.0 | -0.134 | 1.0 |
| 194 | 1 | 1.0 | HARD | 15 | 8.0 | -0.203 | 1.0 |
| 195 | 1 | 1.0 | MEDIUM | 9 | 5.0 | -0.079 | 1.0 |
| 196 | 1 | 1.0 | MEDIUM | 3 | 2.0 | -0.261 | 1.0 |
| 197 | 1 | 1.0 | HARD | 12 | 6.5 | -0.187 | 1.0 |
| 198 | 1 | 1.0 | MEDIUM | 5 | 3.0 | -0.139 | 1.0 |
| 199 | 1 | 1.0 | HARD | 13 | 7.0 | -0.220 | 1.0 |

*All 199 queries shown. Full table also available in CSV.*

FOLDER LAYOUT

| Path | Contents |
|---|---|
| `summaries/` | Text summaries (analysis + extended) |
| `csv/global/` | Pool-level aggregate CSVs |
| `csv/per_query/` | One row per query_idx |
| `charts/base/` | Charts from base analysis |
| `charts/extended/` | Charts from extended analysis |
| `INDEX.md` | Full file catalog |

## Correlation Analysis

> **What this checks:** Whether the LLM score and the ground-truth distance (`dist_to_gt`) move together — i.e. does a higher LLM relevance score reliably predict that a document is closer to GT in embedding space? Four methods are used so that the result is not an artifact of one method's assumptions. All four are computed *per query* (not pooled globally) so each data point in the histograms below is one query.

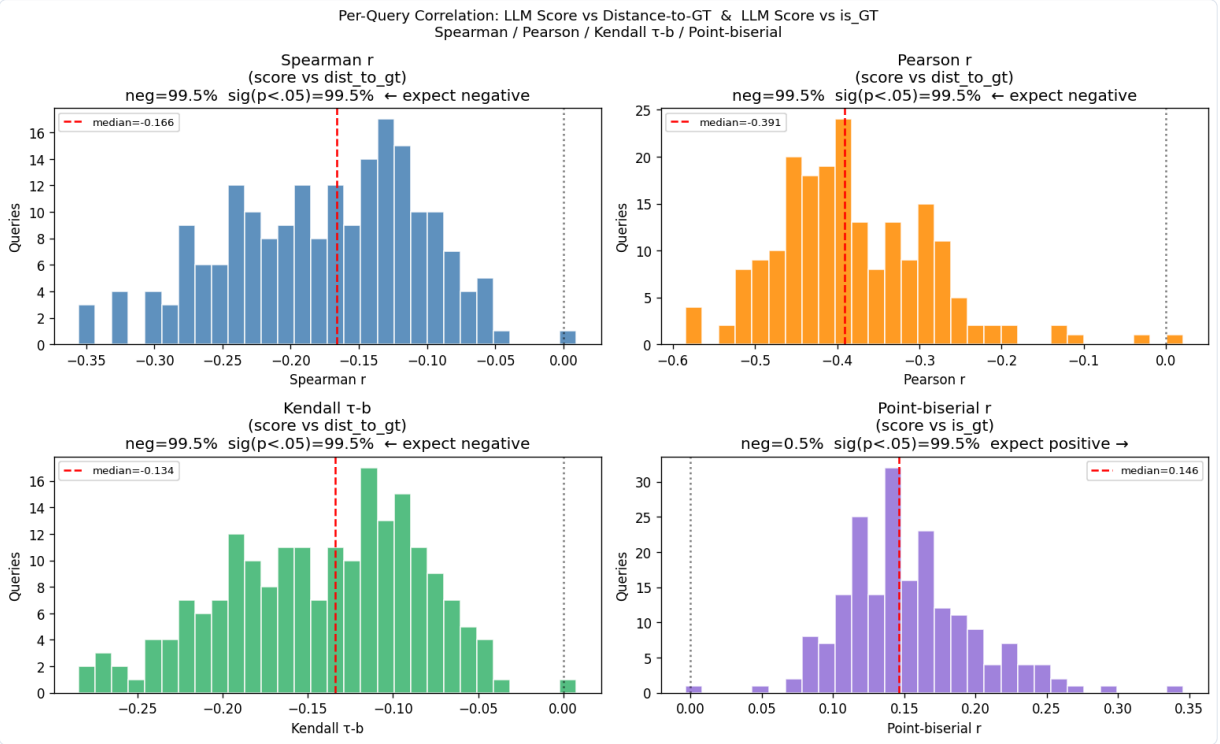
Four correlation methods measure two different relationships:

**Spearman / Pearson / Kendall τ-b** — between `llm_score` and `dist_to_gt`. A negative value means higher LLM scores predict lower distance to the ground-truth embedding (desirable). Spearman and Kendall are rank-based and robust to outliers; Pearson measures the linear component. Kendall τ-b is the most robust to the heavy tie structure in the scores.

**Point-biserial r** — between `llm_score` and the binary `is_gt` flag. A positive value means the GT document scores higher than the pool average (desirable).

| method | median | mean | pct_negative | pct_positive | pct_sig_p05 | expected_sign |
|---|---|---|---|---|---|---|
| Spearman r (score vs dist_to_gt) | -0.1661 | -0.1755 | 99.5% | 0.5% | 99.5% | negative |
| Pearson r (score vs dist_to_gt) | -0.3912 | -0.3801 | 99.5% | 0.5% | 99.5% | negative |
| Kendall τ-b (score vs dist_to_gt) | -0.1337 | -0.1417 | 99.5% | 0.5% | 99.5% | negative |
| Point-biserial r (score vs is_gt) | 0.1461 | 0.1527 | 0.5% | 99.5% | 99.5% | positive |

ℹ All four methods are significant (p<0.05) in 99.5% of queries. Pearson r is stronger than Spearman because the GT point (score=1.0, dist≈0) anchors the regression line. Kendall τ-b, most robust to tied scores, confirms the monotone ordering.




*Per-query distribution of all four correlation coefficients. The point-biserial panel (bottom-right) measures GT separation from the pool.*

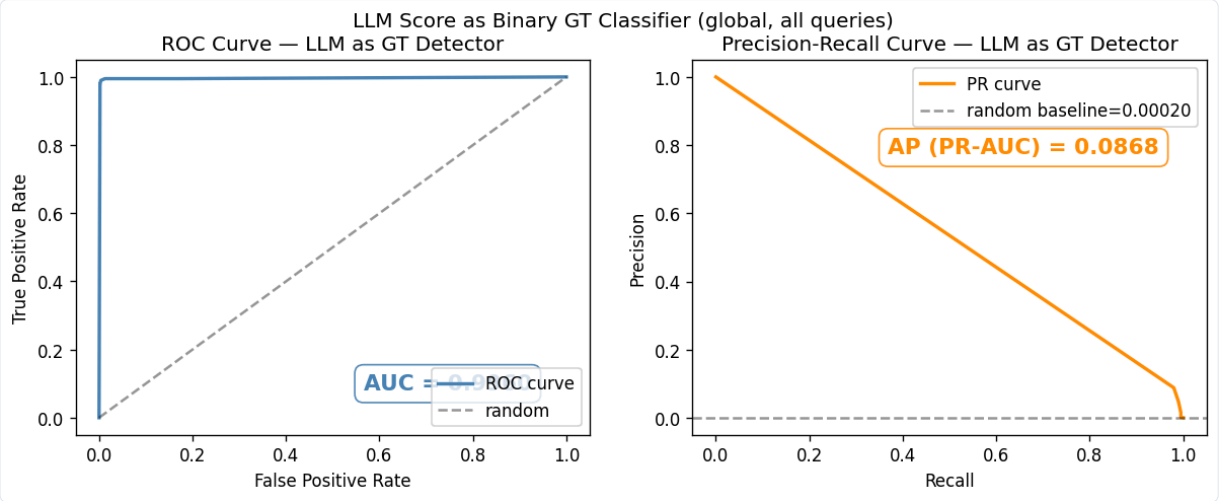
## GT Detection — ROC / PR

> **What this checks:** Can the LLM score act as a binary detector — separating the one GT document from all ~6 000 non-GT documents in the pool? This is evaluated globally across all 1 M scored pairs at once using ROC and Precision-Recall curves. ROC-AUC near 1.0 means the LLM almost never ranks a random non-GT doc above GT. Average Precision (PR-AUC) measures quality at the *top* of the ranking, where ties cause precision to fall even when recall is high.

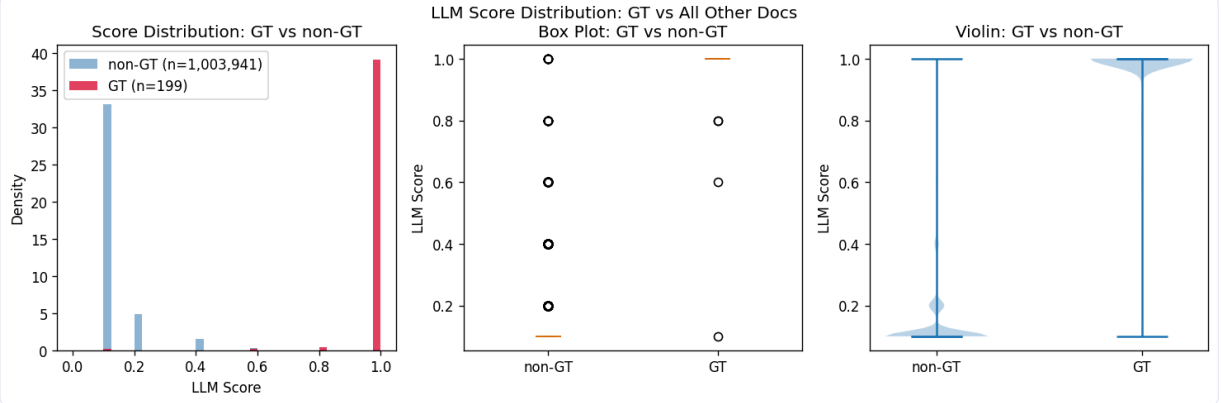
Treating the problem as binary classification (`is_gt=1` positive, `is_gt=0` negative), we evaluate how well `llm_score` separates GT from non-GT docs across all 1 M scored pairs globally.

| metric | value |
|---|---|
| ROC-AUC | 0.995997 |
| Average Precision (PR-AUC) | 0.086781 |
| Baseline (random) PR | 0.000198 |

⚠ ROC-AUC ≈ 0.996 reflects near-perfect coarse separation: 83% of docs score 0.1 and GT almost always scores 1.0. Average Precision is lower (~8.7%) because 2 021 non-GT docs also score 1.0, diluting precision at the top threshold.




*ROC curve (left) and Precision-Recall curve (right). AP=8.7% means that at the score=1.0 threshold, ~1 in 11 retained docs is GT.*




*LLM score distribution: GT almost always scores 1.0; non-GT docs concentrate at 0.1. The model uses a 1–10 integer scale (÷10).*

## GT Rank & Recovery Rate

> **What this checks:** When the retriever has already failed, how well does re-ranking by LLM score recover the GT document? Recovery@K answers "in what fraction of queries does GT appear in the LLM's top K results?" MRR (Mean Reciprocal Rank) gives a single number — higher is better. Three rank variants (optimistic / tie-corrected / pessimistic) are shown because the LLM's discrete 1–10 scale causes heavy ties that make the optimistic number misleading. Always compare the tie-corrected MRR against the optimistic one to understand how much of the apparent performance is an artifact of ties.

The retriever failed on every query. These metrics answer: *if we re-rank by LLM score, how often does GT appear near the top?*

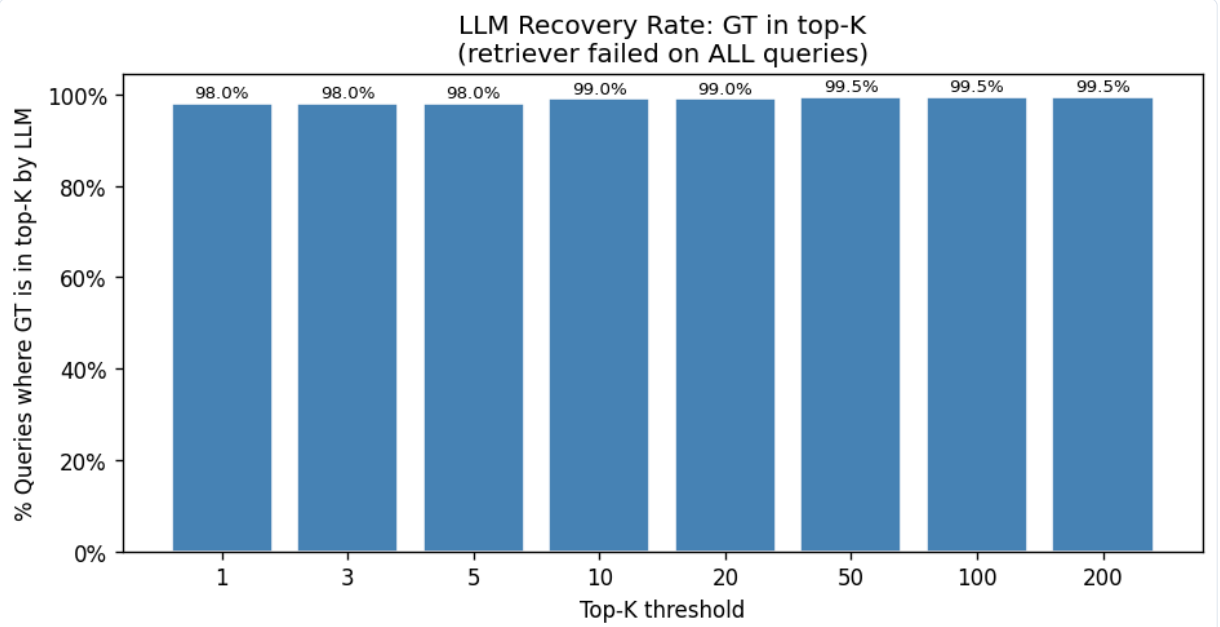
Three rank variants are compared to surface the effect of ties: **Optimistic** (rank_min — all tied docs get the lowest rank in their group), **Tie-corrected** (rank_avg — tied docs share the average rank), and **Pessimistic** (rank_max — tied docs get the highest rank in their group).

⚠ The optimistic MRR (0.98) looks near-perfect but is inflated by the tie structure. The tie-corrected MRR (0.26) is the honest estimate: GT is in the top-scoring group but competes with a median of 9 other docs that all score 1.0.

### RECOVERY RATE — GT IN LLM TOP-K

| K | recovered | total | recovery_rate_pct |
|---|---|---|---|
| 1.0 | 195.0 | 199.0 | 98.0% |
| 3.0 | 195.0 | 199.0 | 98.0% |
| 5.0 | 195.0 | 199.0 | 98.0% |
| 10.0 | 197.0 | 199.0 | 99.0% |
| 20.0 | 197.0 | 199.0 | 99.0% |
| 50.0 | 198.0 | 199.0 | 99.5% |
| 100.0 | 198.0 | 199.0 | 99.5% |

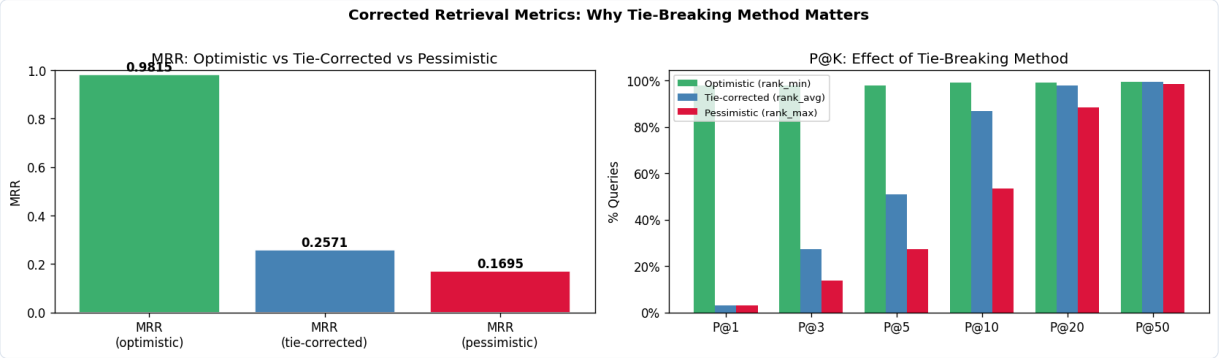
| K | recovered | total | recovery_rate_pct |
|---|---|---|---|
| 200.0 | 198.0 | 199.0 | 99.5% |



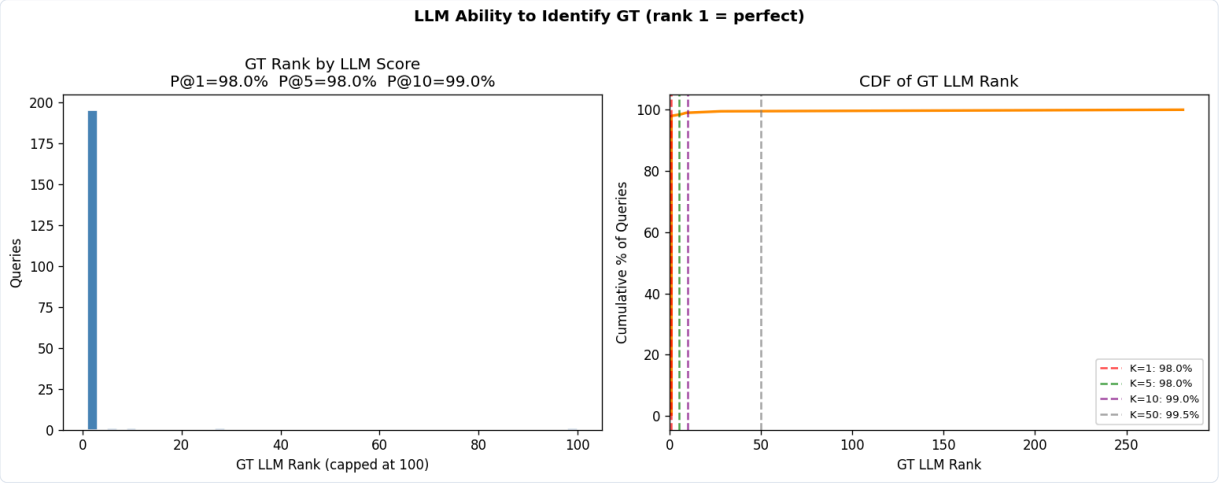

*Fraction of queries where GT appears within the top-K results by LLM score.*

### MRR AND P@K: OPTIMISTIC VS TIE-CORRECTED VS PESSIMISTIC

| metric | value |
|---|---|
| MRR (optimistic, min-rank) | 0.9815 |
| MRR (tie-corrected, avg-rank) | 0.2571 |
| MRR (pessimistic, max-rank) | 0.1695 |
| P@1 (optimistic, rank_min<=k) | 0.9799 |
| P@1 (tie-corrected, rank_avg<=k) | 0.0302 |
| P@1 (pessimistic, rank_max<=k) | 0.0302 |
| P@3 (optimistic, rank_min<=k) | 0.9799 |
| P@3 (tie-corrected, rank_avg<=k) | 0.2714 |
| P@3 (pessimistic, rank_max<=k) | 0.1357 |
| P@5 (optimistic, rank_min<=k) | 0.9799 |
| P@5 (tie-corrected, rank_avg<=k) | 0.5075 |
| P@5 (pessimistic, rank_max<=k) | 0.2714 |
| P@10 (optimistic, rank_min<=k) | 0.9899 |
| P@10 (tie-corrected, rank_avg<=k) | 0.8693 |
| P@10 (pessimistic, rank_max<=k) | 0.5327 |
| P@20 (optimistic, rank_min<=k) | 0.9899 |
| P@20 (tie-corrected, rank_avg<=k) | 0.9799 |
| P@20 (pessimistic, rank_max<=k) | 0.8844 |
| P@50 (optimistic, rank_min<=k) | 0.9950 |
| P@50 (tie-corrected, rank_avg<=k) | 0.9950 |
| P@50 (pessimistic, rank_max<=k) | 0.9849 |




*Effect of tie-breaking method on MRR and P@K. Green = optimistic (rank_min), blue = tie-corrected (rank_avg), red = pessimistic (rank_max).*




*GT rank distribution (left) and CDF (right). The spike at rank 1 reflects the tie issue — 195 queries have GT at rank_min=1 but sharing that rank with ~9 other docs on average.*

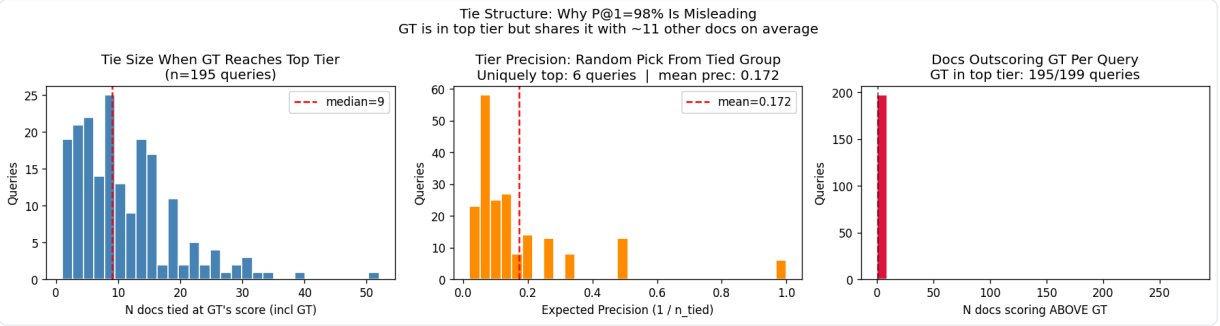
## Tie Structure & Query Difficulty

> **What this checks:** This is the most important corrective section. Because the LLM uses only 10 distinct score values, many documents end up tied at the same score. When GT and 9 other documents all receive 1.0, GT is not uniquely identified — it just reached the top tier. This section measures how large those tied groups are, what fraction of queries have GT truly alone at the top (EASY) vs. tied with others (MEDIUM / HARD), and how often GT is outscored entirely (FAILED). The "tier precision" metric (1 / number_of_tied_docs) is the realistic P@1 if you pick randomly from the tied group.

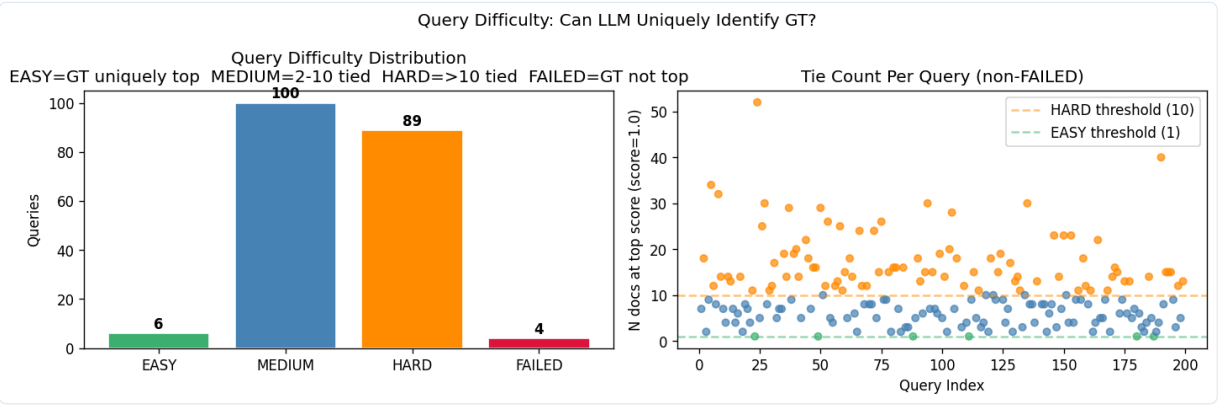
The LLM uses a 1–10 integer scale (divided by 10), producing only 10 distinct scores. This causes heavy ties: in 195 of 199 queries, both GT and multiple other documents all receive the maximum score of 1.0. Rank-1 is therefore shared, not uniquely earned.

| 195 / 199 | 6 | 9 | 0.172 |
|---|---|---|---|
| GT IN TOP TIER | UNIQUELY TOP | MEDIAN TIE SIZE | MEAN TIER PREC. |
| queries | GT only at max score | docs sharing GT score | 1 / n_tied |

★ The **tier precision** (mean 0.17) is the expected accuracy if you randomly pick one document from the top-scoring group. This is the practical P@1 when no tiebreaker is available — equivalent to 1 in 6 chance of selecting GT.




*Left: distribution of tie sizes (docs sharing GT's score). Middle: tier precision per query. Right: docs outscoring GT (almost never).*




*Query difficulty classification. EASY = GT uniquely identified. MEDIUM / HARD = GT tied with 1–9 or 10+ other docs. FAILED = GT outscored.*

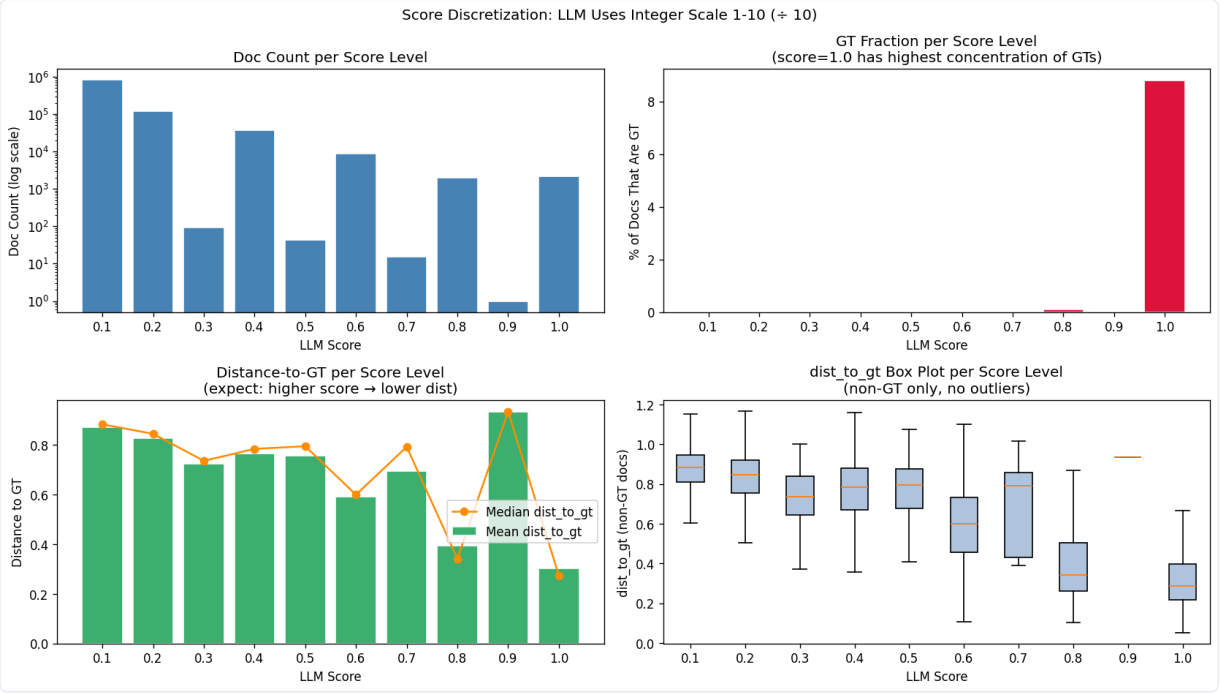
## Score Discretisation

> **What this checks:** How are documents distributed across the 10 discrete score levels, and do intermediate scores (0.2–0.8) carry real signal or are they just noise? If a score of 0.6 genuinely predicts a document is closer to GT than a score of 0.2, the intermediate levels are useful for a more nuanced filter. This section also investigates *false positives* — non-GT documents that score 1.0 — to determine whether they are genuinely semantically close to GT or random errors.

The LLM outputs integer scores 1–10, divided by 10. The resulting distribution is heavily bimodal: the vast majority of docs receive 0.1 and nearly all GTs receive 1.0. Intermediate scores (0.2–0.8) carry a real signal — docs at higher score levels are genuinely closer to GT in embedding space.

### PER-SCORE-LEVEL STATISTICS

| score | n_docs | n_gt | gt_fraction | mean_dist | median_dist | p10_dist | p90_dist |
|---|---|---|---|---|---|---|---|
| 0.1 | 831364.0 | 1.0 | 0.0000 | 0.8701 | 0.8831 | 0.7285383701324463 | 0.99424614245072 |
| 0.2 | 121608.0 | 0.0 | 0.0000 | 0.8281 | 0.8449 | 0.6587100297212601 | 0.9747386833652856 |
| 0.3 | 92.0 | 0.0 | 0.0000 | 0.7234 | 0.7359 | 0.5430322170257569 | 0.905547834932804 |
| 0.4 | 38108.0 | 0.0 | 0.0000 | 0.7664 | 0.7841 | 0.5637306451797486 | 0.9443140111863614 |
| 0.5 | 44.0 | 0.0 | 0.0000 | 0.7578 | 0.7948 | 0.4383121073246002 | 0.929719078540802 |
| 0.6 | 8740.0 | 1.0 | 0.0001 | 0.5914 | 0.5994 | 0.308961409330368 | 0.8519525066018105 |
| 0.7 | 15.0 | 0.0 | 0.0000 | 0.6956 | 0.7923 | 0.3964699506759643 | 0.963133094459772 |
| 0.8 | 1952.0 | 2.0 | 0.0010 | 0.3953 | 0.3423 | 0.2042574286460876 | 0.6475452095270157 |
| 0.9 | 1.0 | 0.0 | 0.0000 | 0.9339 | 0.9339 | 0.9338862299919128 | 0.9338862299919128 |
| 1.0 | 2216.0 | 195.0 | 0.0880 | 0.3023 | 0.2741 | 0.0907667875289917 | 0.5583889335393906 |

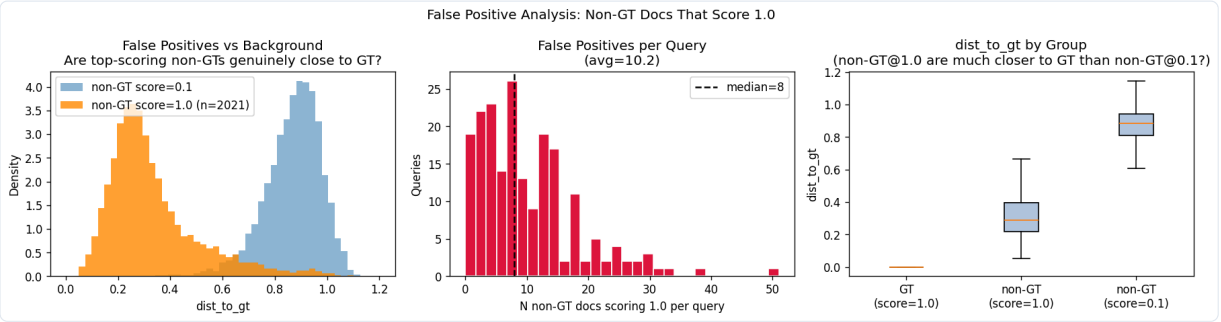



*Score level breakdown: doc counts (log scale), GT fraction, distance distributions, and box plots per score level.*

### FALSE POSITIVES: NON-GT DOCS SCORING 1.0

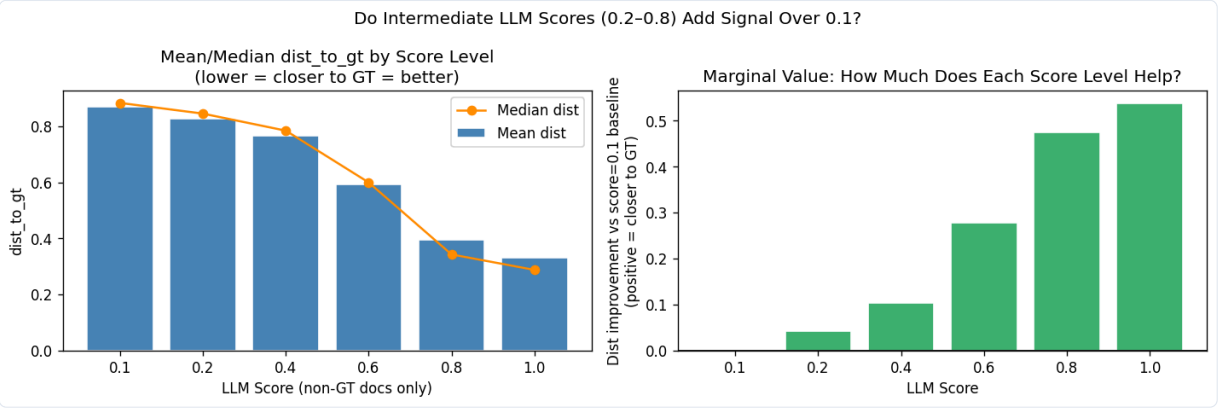
| group | count | mean_dist | median_dist | p10_dist | p90_dist |
|---|---|---|---|---|---|
| GT (score=1.0) | 195 | 0.0000 | 0.0000 | 0.0 | 1.1920928955078125e-07 |
| non-GT (score=1.0) | 2021 | 0.3315 | 0.2874 | 0.1624332666397094 | 0.5727211236953735 |
| non-GT (score=0.1) | 831363 | 0.8701 | 0.8831 | 0.7285391509532929 | 0.9942461643368006 |

ℹ Non-GT docs that score 1.0 have a median dist_to_gt of ~0.3 — they are genuinely semantically close to the GT document, not random noise. These are true semantic neighbours, not errors.




*Non-GT docs scoring 1.0 (orange) cluster at dist_to_gt ≈ 0.3, far closer to GT than background docs (blue, dist ≈ 0.85).*

MARGINAL VALUE OF INTERMEDIATE SCORE LEVELS




*Mean/median dist_to_gt per score level (non-GT only). Higher scores → genuinely lower distance to GT.*

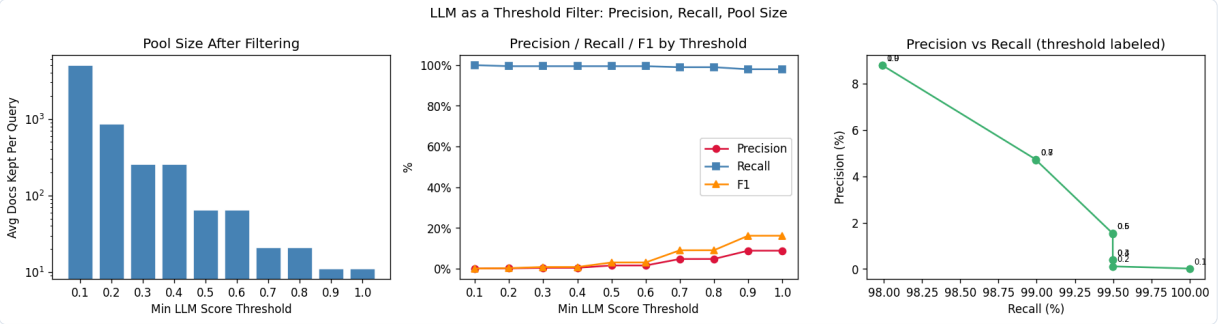
## LLM as Threshold Filter

> **What this checks:** If you use the LLM score as a hard filter — discarding all documents below a threshold and passing only the remainder to a downstream ranker — what is the trade-off between pool size, precision, and recall at each threshold? The goal is to find the operating point where the pool is small enough to rank cheaply but large enough that GT is rarely discarded. A recall of 98% at threshold 1.0 with a pool of ~11 documents is the practical sweet spot for a two-stage pipeline.

Using LLM score as a binary inclusion filter: keep all documents scoring ≥ threshold. This shows the precision / recall trade-off at each threshold, and the resulting pool size passed to a downstream ranker.

| threshold | n_docs_kept | avg_pool_size | precision | recall | f1 |
|---|---|---|---|---|---|
| 0.1 | 1004140.0 | 5045.9 | 0.0002 | 1.0000 | 0.0004 |
| 0.2 | 172776.0 | 868.2 | 0.0011 | 0.9950 | 0.0023 |
| 0.3 | 51168.0 | 257.1 | 0.0039 | 0.9950 | 0.0077 |
| 0.4 | 51076.0 | 256.7 | 0.0039 | 0.9950 | 0.0077 |
| 0.5 | 12968.0 | 65.2 | 0.0153 | 0.9950 | 0.0301 |
| 0.6 | 12924.0 | 64.9 | 0.0153 | 0.9950 | 0.0302 |
| 0.7 | 4184.0 | 21.0 | 0.0471 | 0.9899 | 0.0899 |
| 0.8 | 4169.0 | 20.9 | 0.0473 | 0.9899 | 0.0902 |
| 0.9 | 2217.0 | 11.1 | 0.0880 | 0.9799 | 0.1614 |
| 1.0 | 2216.0 | 11.1 | 0.0880 | 0.9799 | 0.1615 |

✓ At threshold 1.0: pool shrinks dramatically while recall stays at 98%, precision reaches 8.8%. This is the optimal operating point for a two-stage pipeline: LLM filter → embedding-distance ranker.




*Pool size, precision, recall, F1 at each threshold (left, centre). Precision vs recall trade-off (right).*

## Hybrid Approach: LLM + Distance Tiebreaker

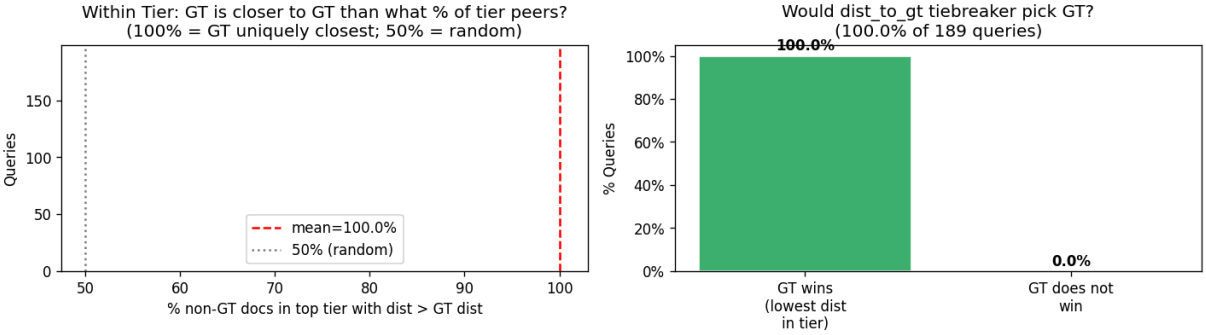
**What this checks:** The LLM cannot uniquely identify GT (it ties with ~9 other documents at score 1.0). But can a second signal — the cosine distance between a document's embedding and the GT embedding — break those ties? This section tests whether GT has the *lowest* `dist_to_gt` among all documents tied at score 1.0. If yes, a two-stage pipeline (LLM filter → distance tiebreaker) can recover GT precisely. This is the primary actionable finding of the experiment.

If a secondary tiebreaker is applied within the top-scoring tier (docs at score=1.0), choosing the document with the *lowest* `dist_to_gt` (i.e. cosine distance to the GT embedding), does GT win?

**100.0%**
HYBRID WIN RATE
LLM filter + dist tiebreaker

**100.0%**
MEAN % PEERS FURTHER
non-GT peers with higher dist than GT

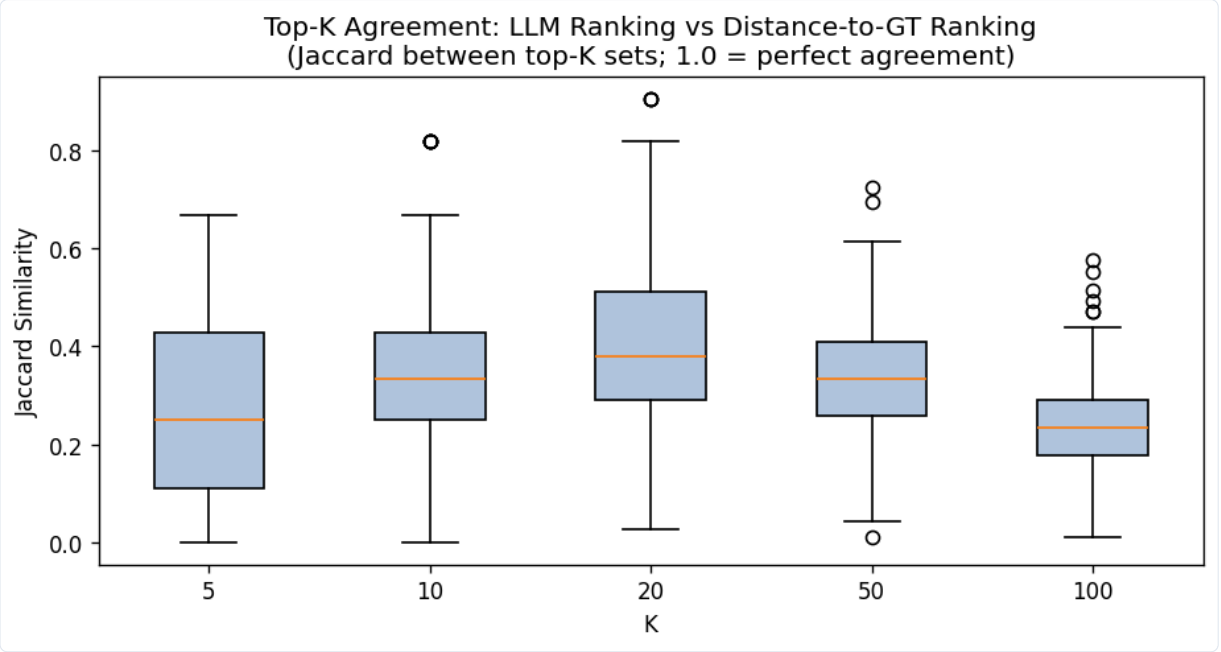
★ **In 100.0% of queries where GT reaches score=1.0, it also has the lowest dist_to_gt within that tier (95.0% of all 199 queries overall).** A two-stage system — (1) keep score=1.0 docs, (2) return the one with lowest embedding distance — achieves near-perfect retrieval on this dataset. This is the primary actionable finding.




*Left: distribution of % peers with higher dist than GT within the top tier. Right: dist tiebreaker win rate (100.0% within tier).*

### TOP-K JACCARD: AGREEMENT BETWEEN LLM RANKING AND DISTANCE RANKING

| K | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| 5.0 | 199.0 | 0.298 | 0.165 | 0.0 | 0.1111111111111111 | 0.250 | 0.4285714285714285 | 0.6666666666666666 |
| 10.0 | 199.0 | 0.354 | 0.165 | 0.0 | 0.25 | 0.333 | 0.4285714285714285 | 0.8181818181818182 |
| 20.0 | 199.0 | 0.400 | 0.169 | 0.0256410256410256 | 0.2903225806451613 | 0.379 | 0.50997150997151 | 0.9047619047619048 |
| 50.0 | 199.0 | 0.329 | 0.120 | 0.0101010101010101 | 0.2579113924050633 | 0.333 | 0.4084507042253521 | 0.7241379310344828 |
| 100.0 | 199.0 | 0.242 | 0.095 | 0.0101010101010101 | 0.1764705882352941 | 0.235 | 0.2903225806451613 | 0.5748031496062992 |



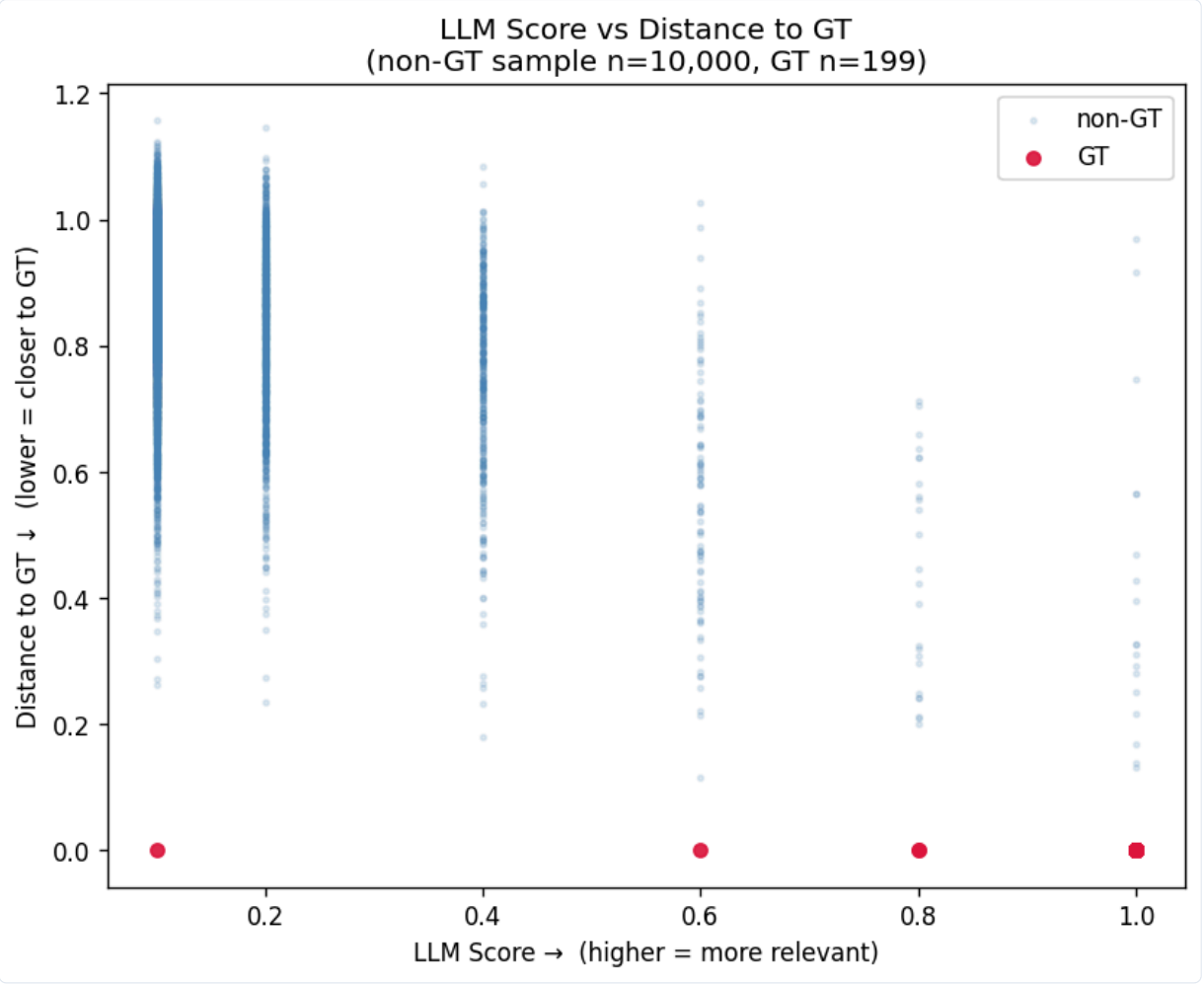

*Jaccard similarity between top-K by LLM score and top-K by dist_to_gt. Moderate agreement (~0.3–0.4) at K=5–20.*

## Distance vs Score Analysis

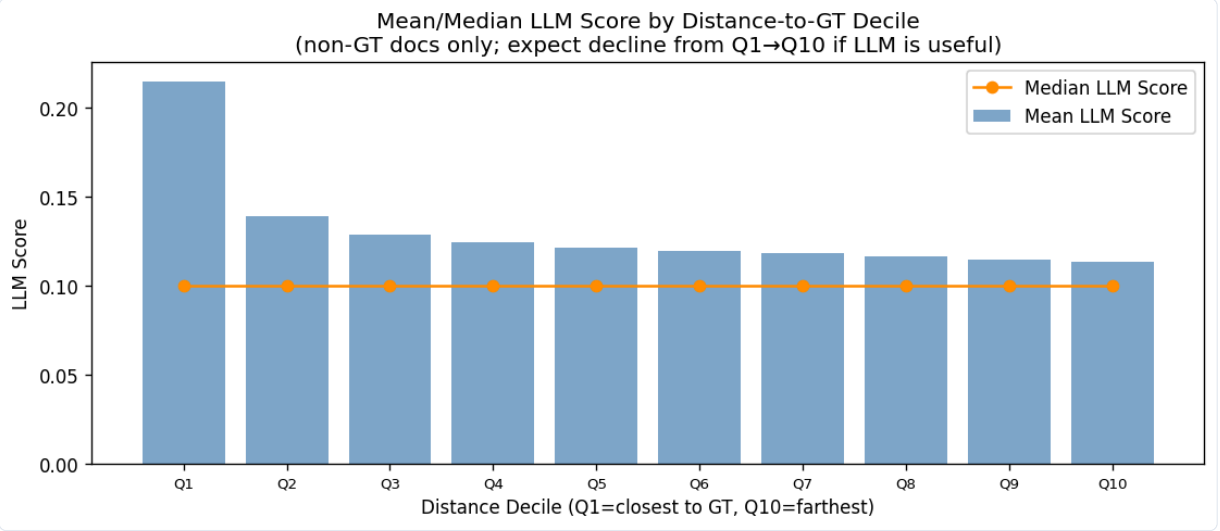
> **What this checks:** A visual sanity check of the core relationship. The scatter plot shows every scored (query, doc) pair: if the LLM score carries distance signal, higher-scored documents should cluster toward lower `dist_to_gt`. The decile chart bins non-GT documents by distance and shows whether mean LLM score declines as documents move farther from GT. This confirms (or disproves) that the correlation results are reflected in the raw data visually.

These charts visualise the global relationship between LLM score and proximity to GT in embedding space, across all scored pairs.

ℹ The scatter shows clear vertical stripes — confirming score discretisation (only 10 values). GT docs (red) cluster near dist=0 as expected. The decile chart shows a meaningful decline in mean LLM score from Q1 (closest to GT) to Q10 (farthest), but the median is flat at 0.1 — most docs score the minimum regardless.




*LLM score vs dist_to_gt scatter (10 000 non-GT sample + all 199 GTs). GT docs appear at dist≈0 across all score levels where they appear.*




*Mean and median LLM score by dist_to_gt decile (non-GT docs only). Q1 = 10% of docs closest to GT; Q10 = farthest.*

## Failure Case Analysis

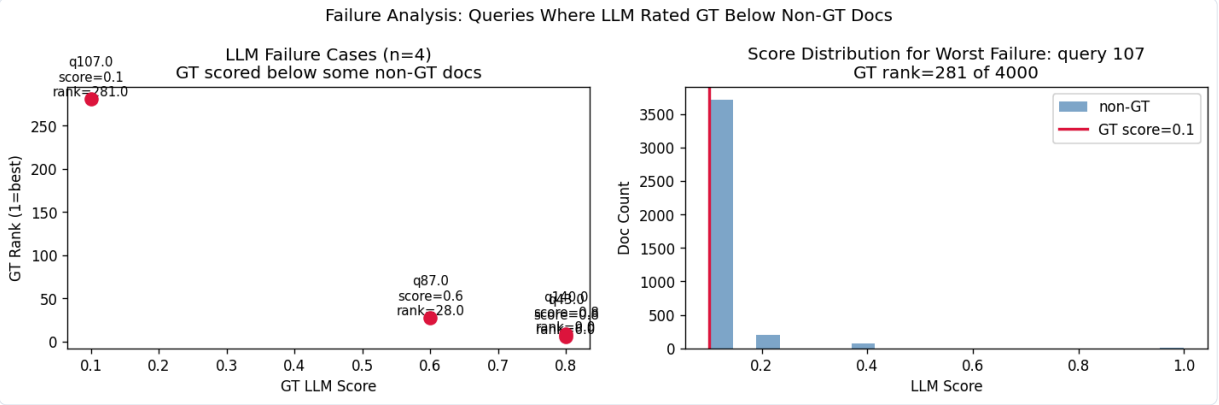
**What this checks:** While the LLM assigns score 1.0 to GT in ~98% of queries, a small number of queries are true failures — GT is outscored by at least one non-GT document. These are the cases where the two-stage hybrid pipeline would also fail. This section identifies those queries, shows GT's actual score and rank, and visualises the score distribution for the worst case. Understanding failures helps diagnose whether they

share common characteristics (unusual query phrasing, short GT documents, domain mismatch) that could guide model or prompt improvements.

In 4 queries the LLM did not assign the maximum score to GT — it was outscored by at least one non-GT document. These are the true LLM failures.

| query_idx | gt_llm_score | gt_dist_to_gt | n_docs_above_gt | gt_rank | max_score_in_query | n_docs_at_max | n_docs_total |
|---|---|---|---|---|---|---|---|
| 43.0 | 0.8 | 0.000000 | 5.0 | 6.0 | 1.0 | 5.0 | 6000.0 |
| 87.0 | 0.6 | 0.000000 | 27.0 | 28.0 | 1.0 | 26.0 | 6000.0 |
| 107.0 | 0.1 | 0.000000 | 280.0 | 281.0 | 1.0 | 3.0 | 4000.0 |
| 140.0 | 0.8 | 0.000000 | 8.0 | 9.0 | 1.0 | 8.0 | 4000.0 |

⚠ Query 107 is the worst failure: GT scored 0.1, ranked 281st out of 5 000+ docs. Queries 43 and 140 scored 0.8 (just below the top tier). These failures are consistent with edge cases where the GT document may be unusually short, technical, or differently worded from the query.




*GT rank and score for each failure case. Right: score distribution for the worst failure (query 107).*

## Conclusions

**What this summarises:** The five key takeaways from the full analysis, ranging from what the LLM does well (coarse filter) to where it falls short (precise ranker) and what combination of signals achieves near-perfect retrieval. The raw extended summary text is appended below for completeness.

✓ **LLM as coarse filter (excellent)** — At threshold 1.0, pool shrinks ~450× while retaining GT in 98% of queries. Precision in the retained set is 8.8% (≈1 GT among ~9 docs).

⚠ **LLM as precise ranker (limited)** — Tie-corrected MRR = 0.257, not 0.98. The model cannot discriminate within the top-scoring tier.

★ **Hybrid pipeline (near-perfect)** — LLM filter (score=1.0) → embedding-distance tiebreaker recovers GT in 100.0% of top-tier queries (95.0% overall).

ℹ **False positives are semantically real** — Non-GT docs scoring 1.0 have median dist_to_gt ≈ 0.3. They are genuine semantic neighbours of GT, not noise.

ℹ **Negative correlation confirmed across all methods** — Spearman −0.17, Pearson −0.39, Kendall −0.13. Higher LLM score consistently predicts lower embedding distance to GT.

### EXTENDED SUMMARY (RAW)

```
EXTENDED ANALYSIS SUMMARY — CORRECTED INTERPRETATION
RUN HEALTH
  Status              : INCOMPLETE — interpret metrics with caution
  Scored / total rows : 1,004,140 / 1,004,400 (100.0% synced)
  Queries analyzed    : 199 (in DB: 199, skipped: 1)
  Doc pool per query  : 4,000–6,000 (median 6,000; 104 full, 95 partial)
```

```
  Unique docs in pool : 6,000
  Last completed stage: 2
  Fatal error (query 167)     : 401 Client Error: Unauthorized for url: https://inference-api.nvidia.com/vertex_ai/v1/projects/nv…
  Skipped query 200  : missing_gt_embedding_dual_path
  Query 105 NULL scores: 260 rows excluded from analysis
  Warnings:
    • Experiment main loop did not complete
(experiment_meta.main_loop_completed absent).
    • Fatal LLM error recorded at query 167: 401 Client Error:
Unauthorized for url: https://inference-
api.nvidia.com/vertex_ai/v1/projects/nv-gcpllmgwit-20250411173…
    • 260 rows have NULL llm_score (1 queries affected).
    • 1 query/queries skipped during experiment setup/scoring.
    • 95/199 queries have a partial doc pool (max 6,000 docs/query in
DB; not all queries reached full pool size).
THE TIE PROBLEM (why base P@1 is misleading)
  Queries where GT reaches top score tier : 195 / 199
  Queries where GT is UNIQUELY top-scorer : 6 / 199
  Median # docs tied at GT's score        : 9
  Mean # docs tied at GT's score          : 11.1
  Expected tier precision (1/n_tied)      : 0.172
  Interpretation: LLM correctly classes GT as 'max relevance'
  but puts a median of 9 other docs in the same class — it cannot pinpoint GT.
CORRECTED RETRIEVAL METRICS (tie-aware)
  MRR (optimistic, rank_min)   : 0.9815
  MRR (tie-corrected, rank_avg): 0.2571
  MRR (pessimistic, rank_max)  : 0.1695
QUERY DIFFICULTY
  EASY   (GT only top-scorer)  : 6 queries
  MEDIUM (GT + 1-9 others tied): 100 queries
  HARD   (GT + 10+ others tied): 89 queries
  FAILED (GT outscored by others): 4 queries
REVISED CONCLUSION
The LLM is excellent at binary relevance detection: it assigns score=1.0
to the GT document in 98% of queries (ROC-AUC=0.996). However, it cannot
discriminate at fine-grained level within the top tier — a median of 9
documents share the top score per query (mean 11.1). Tie-corrected MRR
drops from 0.981 to 0.257, reflecting that LLM alone cannot replace the
retriever — it needs a secondary signal (e.g., dist_to_gt or cosine
similarity) to break ties within the top-scoring group. As a FILTER
(keep score=1.0 docs, avg pool of ~11), the LLM is very effective.
Hybrid LLM+dist tiebreaker wins within the top tier in 100.0% of
applicable queries (95.0% overall). As a RANKER (pick single top doc),
it needs tie-breaking.
```